

2. Conectaros con www.fib.upc.edu. En esta conexión os enviarán el certificado del servidor de la FIB.

El paquete Client Hello de www.fib.upc.edu es el número 301.

El paquete Server Hello de www.fib.upc.edu es el número 307.

El paquete Certificate, Server de www.fib.upc.edu es el número 319.

El fichero para descifrar los paquetes es TLS_keys_descifrar_fib.txt

299	9.177360777	192.168.1.16	147.83.249.103	TCP	68 36904 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=83209
300	9.179794402	192.168.1.16	147.83.249.103	TCP	1516 36904 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=1448 TSval=83209
301	9.179815232	192.168.1.16	147.83.249.103	TLSv1.2	520 Client Hello
306	9.186067835	147.83.249.103	192.168.1.16	TCP	68 443 → 36904 [ACK] Seq=1 Ack=1901 Win=32768 Len=0 TSval=2050
307	9.186068092	147.83.249.103	192.168.1.16	TLSv1.2	4164 Server Hello
308	9.186122064	192.168.1.16	147.83.249.103	TCP	68 36904 → 443 [ACK] Seq=1901 Ack=4097 Win=60160 Len=0 TSval=2050
319	9.201252450	147.83.249.103	192.168.1.16	TLSv1.2	1628 Certificate, Server Key Exchange, Server Hello Done
320	9.201288518	192.168.1.16	147.83.249.103	TCP	68 36904 → 443 [ACK] Seq=1901 Ack=5657 Win=58624 Len=0 TSval=2050
324	9.206545909	192.168.1.16	147.83.249.103	TLSv1.2	194 Client Key Exchange, Change Cipher Spec, Encrypted Handshake
327	9.213490157	147.83.249.103	192.168.1.16	TLSv1.2	342 New Session Ticket, Change Cipher Spec, Encrypted Handshake
329	9.214385199	192.168.1.16	147.83.249.103	TLSv1.2	544 Application Data

(a) Obtened el periodo de validez del certificado y la clave pública (RSA: módulo y exponente; ECC: curva y coordenadas X e Y del punto) del web de la FIB.

El periodo de validez del certificado es desde UTC 2025-11-21 09:18:51 (not before) hasta UTC 2026-11-21 09:18:51 (not after) en.

▼ Certificate: 308207063082056ea0030201020210540cb6b83dbb8ced1ba201e50fc
▼ signedCertificate
version: v3 (2)
serialNumber: 0x540cb6b83dbb8ced1ba201e50fc74bf
▶ signature (sha256WithRSAEncryption)
▶ issuer: rdnSequence (0)
▼ validity
▼ notBefore: utcTime (0)
utcTime: 2025-11-21 09:18:51 (UTC)
▼ notAfter: utcTime (0)
utcTime: 2026-11-21 09:18:51 (UTC)
▼ subject: rdnSequence (0)
▶ rdnSequence: 1 item (id-at-commonName=www.fib.upc.edu)
▶ subjectPublicKeyInfo
▶ extensions: 9 items
▶ algorithmIdentifier (sha256WithRSAEncryption)

fib.upc.edu utiliza el algoritmo RSA. La clave pública se encuentra en subjectPublicKeyInfo:

módulo=

0x00d8708c71ea35e34ab215f3f876684f588e62790780485aaf06d3aabf2ec8ca2cb727f116aee47a59a4b39034e72f4dee71bb28c5bbd2a4355df70928acba783e3caa62e968bdb805e24b3fe31af9136062c8bbe
c565444e7b4cf5432cb4762c531135a5cac24053ed706208cae9aea8b0a09f32f62f7ba8befca00f41aab6
8aae1392c272e8ca20906cea4d02ff7a5d3d88bf2360c3e6d37cfdc59e42bec35a2b3dd17549f41dc102bb
6518172835840b1019a00c08c4e70daccbc86512a0a214aa1466d44ca994156c45ecfffac3c4b69d41e7b
fb62c0870d24885b5eab7e8a472f3e383b693055fa20291401ccd66ba05d1e7562328f57cf50cfde3d997
992ccf933d03cf7eae271fd137c58bdd67e688b10bf115a523290c6278b3adc4ee22168379bef6063251c
7346107804d1327a63438bafeaf1df4152daab8e30bb73b31d0638ec1f46d68611bd8611e584861d892c

6d84c969106d2737c7f22efc25063118d8b95681f35accd73f06cf52f9e762e0942e3722708a9a0bc4d8d4910d

exponente público = 65537

```
    subjectPublicKeyInfo
      algorithm (rsaEncryption)
      subjectPublicKey: 3082018a0282018100d8708c71ea35e34ab215f3f87668
        modulus: 0x00d8708c71ea35e34ab215f3f876684f588e62790780485aaf0
        publicExponent: 65537
```

(b) Indicad las marcas de tiempo de los certificados, hasta el milisegundo (Signed Certificate Timestamp).

En el campo Extension (SignedCertificateTimestampList) del certificado de la FIB:

Nov 21, 2025 09:28:57.277000000 UTC

Nov 21, 2025 09:28:57.228000000 UTC

Nov 21, 2025 09:28:57.281000000 UTC

```
    Extension (id-ce-cRLDistributionPoints)
    Extension (id-ce-subjectKeyIdentifier)
    Extension (id-ce-keyUsage)
    Extension (SignedCertificateTimestampList)
      Extension Id: 1.3.6.1.4.1.11129.2.4.2 (SignedCertificate
      Serialized SCT List Length: 360
      Signed Certificate Timestamp (Unknown Log)
        Serialized SCT Length: 117
        SCT Version: 0
        Log ID: 944e4387faecc1ef81f3192426a8186501c7d35f380201
        Timestamp: Nov 21, 2025 09:28:57.277000000 UTC
        Extensions length: 0
      Signature Algorithm: ecdsa_secp256r1_sha256 (0x0403)
      Signature Length: 70
      Signature: 304402203c99eec527d8b496060156e9fd00ecc9462
    Signed Certificate Timestamp (Unknown Log)
    Signed Certificate Timestamp (Unknown Log)
    algorithmIdentifier (sha256WithRSAEncryption)
    Padding: 0
```

(c) En el certificado encontraréis los DNS para los que es válido, indicadlos.

Los DNS para los que el certificado de fib.upc.edu es válido son: www.fib.upc.edu y www.fib.upc.es

```

    ▾ Extension (id-ce-subjectAltName)
      Extension Id: 2.5.29.17 (id-ce-subjectAltName)
      ▾ GeneralNames: 2 items
        ▾ GeneralName: dNSName (2)
          dNSName: www.fib.upc.edu
        ▾ GeneralName: dNSName (2)
          dNSName: www.fib.upc.es

```

(d) En el certificado encontraréis un enlace un punto de distribución de la CRL de la autoridad certificadora. ¿Cuántos certificados revocados contiene la CRL?, ¿cuántos hay sido revocados por Key Compromise?

El enlace al punto de distribución de la CRL es uniformResourceIdentifier:
<http://crl.harica.gr/HARICA-GEANT-TLS-R1.crl>

```

    ▸ subjectPublicKeyInfo
    ▾ extensions: 9 items
      ▸ Extension (id-ce-authorityKeyIdentifier)
      ▸ Extension (id-pe-authorityInfoAccess)
      ▸ Extension (id-ce-subjectAltName)
      ▸ Extension (id-ce-certificatePolicies)
      ▸ Extension (id-ce-extKeyUsage)
      ▾ Extension (id-ce-cRLDistributionPoints)
        Extension Id: 2.5.29.31 (id-ce-cRLDistributionPoints)
        ▾ CRLDistPointsSyntax: 1 item
          ▾ DistributionPoint
            ▾ distributionPoint: fullName (0)
              ▾ fullName: 1 item
                ▾ GeneralName: uniformResourceIdentifier (6)
                  uniformResourceIdentifier: http://crl.harica.gr/HARICA-GEANT-TLS-R1.crl
      ▸ Extension (id-ce-subjectKeyIdentifier)

```

Al ejecutar los comandos

```
# contar numero de certificados revocados
os.system("openssl crl -in HARICA-GEANT-TLS-R1.crl -text -noout | grep 'Revocation Date' | wc -l")
```

```
# contar numero de certificados revocados por Key Compromise
os.system("openssl crl -in HARICA-GEANT-TLS-R1.crl -text -noout | grep 'Key Compromise' | wc -l")
```

obtengo que se han revocado 13220 certificados en total, de los cuales 456 a causa de Key Compromise.

(e) En el certificado encontraréis la dirección OCSP (Online Certificate Status Protocol) a la que

se puede preguntar por el estatus del certificado. ¿Cuál es el estatus del certificado y hasta cuándo es válido dicho estatus?

La dirección OSCP es uniformResourceIdentifier: <http://ocsp-tls.harica.gr>

Ejecutando el comando

```
os.system("openssl ocsp -issuer certificado_CA.pem -cert certificado_FIB.pem -url  
http://ocsp-tls.harica.gr -resp_text -respout status.der")
```

obtengo el estado del certificado de la FIB. El estatus del certificado es good. El estatus es válido hasta Next Update: Dec 13 11:17:01 2025 GMT. El volcado obtenido en el momento de la ejecución (día 10/12/25) del comando es el siguiente:

OCSP Response Data:

```
OCSP Response Status: successful (0x0)  
Response Type: Basic OCSP Response  
Version: 1 (0x0)  
Responder Id: 8601723F8CA970E231065316CE015F5B79C83C3B  
Produced At: Dec 10 11:17:02 2025 GMT  
Responses:  
Certificate ID:  
  Hash Algorithm: sha1  
  Issuer Name Hash: 2892028EABFCA4FFDDA1B5EC1474688142C20012  
  Issuer Key Hash: 8601723F8CA970E231065316CE015F5B79C83C3B  
  Serial Number: 540CB6B83DBB8CEDE1BA201E50FC74BF  
Cert Status: good  
This Update: Dec 10 11:17:02 2025 GMT  
Next Update: Dec 13 11:17:01 2025 GMT
```

Signature Algorithm: sha256WithRSAEncryption

Signature Value:

```
4d:09:a6:07:f6:08:ca:f6:0d:6f:3f:a8:67:9e:ef:53:a0:6a:  
14:65:b1:86:35:16:30:57:f3:df:67:30:ce:24:cc:57:8e:52:  
66:f8:f4:fe:e0:b1:1d:32:7f:35:d7:83:1f:de:12:fe:35:32:  
71:b9:c3:d8:a0:4e:b2:f4:8f:dd:5c:f2:f6:dd:82:42:37:f2:  
ff:a4:2a:ef:b1:23:18:5c:a6:08:05:de:a4:62:4e:ce:73:22:  
08:eb:97:fc:fe:da:ce:55:63:e0:97:19:58:47:6b:88:1b:11:  
df:80:5a:74:81:e2:05:32:de:b6:a0:d4:18:09:c4:34:66:2e:  
d4:ea:ba:18:60:6b:0c:76:98:55:55:5c:59:57:b3:bf:41:20:  
ad:b6:47:00:ad:6f:88:8f:d1:d4:34:83:d3:dd:a4:26:fb:06:  
8f:c1:92:35:2a:71:e1:1e:52:e5:09:7f:81:d6:c0:41:2d:b4:  
17:85:59:b9:25:2e:77:c6:58:fe:fd:39:86:9e:b3:f7:d9:18:  
b9:fc:79:39:b1:9b:ba:69:da:e0:cb:73:51:ac:d6:94:b5:7a:  
ba:31:82:9e:33:96:b3:0b:d0:23:0a:61:a7:54:fb:19:62:ee:  
7d:c3:e5:44:3d:63:5f:23:25:29:9c:e9:ca:67:40:e4:4a:af:  
68:4d:cc:e8:8f:74:99:c5:e6:a0:c5:2a:e6:b6:4c:64:5f:74:  
6f:3a:a9:a1:09:c5:92:bc:6b:20:4a:97:6a:ed:79:0d:33:5d:  
3f:44:6a:b5:6d:70:af:df:30:ac:e8:fd:e8:ee:62:58:3c:ad:
```

0e:2e:60:2f:f6:c8:8d:63:af:16:0d:5a:9f:6b:7c:aa:9e:48:
fe:05:89:59:68:20:22:f4:16:b4:3b:56:82:43:2a:c5:05:72:
09:a4:a4:a8:41:3c:e0:c4:87:7f:ed:8f:51:c6:1a:79:3d:cd:
ed:65:16:59:a8:76:a9:27:99:40:7f:83:6f:a0:c0:ac:94:10:
12:26:4f:48:eb:c3

WARNING: no nonce in response

Response verify OK

certificado_FIB.pem: good

This Update: Dec 10 11:17:02 2025 GMT

Next Update: Dec 13 11:17:01 2025 GMT

```
  ▶ signature (sha256WithRSAEncryption)
  ▶ issuer: rdnSequence (0)
  ▶ validity
  ▶ subject: rdnSequence (0)
  ▶ subjectPublicKeyInfo
  ▼ extensions: 9 items
    ▶ Extension (id-ce-authorityKeyIdentifier)
    ▼ Extension (id-pe-authorityInfoAccess)
      Extension Id: 1.3.6.1.5.5.7.1.1 (id-pe-authorityInfoAccess)
      ▼ AuthorityInfoAccessSyntax: 2 items
        ▶ AccessDescription
        ▼ AccessDescription
          accessMethod: 1.3.6.1.5.5.7.48.1 (id-ad-ocsp)
          ▼ accessLocation: 6
            uniformResourceIdentifier: http://ocsp-tls.harica.gr
        ▶ Extension (id-ce-subjectAltName)
```